Adam Patterson

Economist & Data Science Engineer

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Education

Central Connecticut State University — New Britain, CT
Master of Science in Data Science — *2025*

University of Connecticut — Storrs, CT
Master of Science in Quantitative Economics — *2020*

University of Connecticut — Storrs, CT
Bachelor of Arts in Economics — *2019*

Experience

Amazon — Jun. 2024 – Present
Data Scientist (Economist Apprentice) — *Seattle, WA*

- Developed Surrogate Index Models translating LLM product updates into long-term business impact forecasts
- Reduced production model standard error by 6% via bias-injected simulations
- Built Causal Discovery framework using Peter-Clark (PC) algorithms and DAG inference to identify structural relationships in high-dimensional observational data
- Constructed interactive Agentic Causal Discovery pipeline mapping data inputs to optimal algorithm selection (PC, FCI, GES) via decision rules and LLM reasoning, with human-in-the-loop feedback
- Applied Double Machine Learning (DML) to estimate treatment effects under high-dimensional covariate spaces
- Designed and executed A/B tests to estimate incremental treatment lift and quantify heterogeneous treatment effects, accounting for multiple testing correction and SUTVA violations
- Engineered scalable data pipelines (SQL, Athena, Glue) and automated monthly model runs in AWS via Lambda and SageMaker; built dashboards in QuickSight

University of Connecticut — Dec. 2019 – Present
Research Technician — *Storrs, CT*

- Architected a RAG system for graduate-level econometrics coursework (ECON5501)
- Deployed LLM-based persona agents to pilot survey instruments and identify potential response bias
- Managed end-to-end data collection, processing, analysis, and reporting for 10+ published research studies
- Utilized quasi-experimental designs (DiD, event study, IPTW, IV) to quantify causal impacts on student learning outcomes
- Evaluated heterogeneous treatment effects (HTE) across diverse populations to inform educational policy
- Delivered instruction in econometrics and microeconomics, cultivating analytical and quantitative skills in undergraduate students

Bryant University — Dec. 2023 – May 2024
Adjunct Professor — *Smithfield, RI*

- Instructor of ECON 114 Principles of Macroeconomics, covering GDP, inflation, unemployment, interest rates, international finance, and international trade
- Led projects where students collected, managed, visualized, and analyzed economic data using FRED
- Designed assignments for students to create dashboards analyzing how policy shifts influence key economic indicators

Nichols College — Dec. 2022 – Jul. 2024
Adjunct Professor — *Dudley, MA*

- Taught DS 201 Introduction to Data Science, covering Python programming and predictive modeling for business applications
- Developed students' skills in data management, visualization, and advanced modeling using Excel, PowerBI, and Tableau
- Supervised a team of four undergraduates in assessing VR intervention impact through pre- and post-surveys

Skills

Technical: SQL, Python, R, AWS (Bedrock, S3, Lambda, SageMaker, Athena, Glue, EC2, ECR, Containers, QuickSight), AutoGluon, PowerBI, Tableau, Excel, LangChain, LangGraph
Statistical: Causal Inference, Causal Discovery, Double Machine Learning, Causal Forests, A/B Testing & Experimentation Design, Treatment Effect Estimation, Regression Analysis, Panel Data, Hypothesis Testing, Machine Learning (Supervised/Unsupervised), Deep Learning, Bayesian Hierarchical Modeling, Time Series Analysis
Domain: Econometrics, Data Science, Agentic AI, Large Language Models, Retrieval Augmented Generation, Natural Language Processing, AutoML, Big Data, Data Visualization

Publications

10. Zhao, W., **Patterson, A.**, Sentihilkumar, C., Wang, Y., Gupta, A., Sadighi, S., & Nayyar, N. (2025). "From Metrics to Meaning: Estimating User Feedback Using LLM-Based Evaluation." *Amazon Science*. [link]
9. Frydenberg, M., Mentzer, K., & **Patterson, A.** (2026). "The Rapid Rise of Generative AI Adoption among First-Year College Students." *Information Systems Education Journal*, 24(1). [link]
8. **Patterson, A.** (2025). "Assessing the Alignment of FOMC Statements with Minutes using Large Language Models." *Issues in Information Systems*, 26(2). [link]
7. **Patterson, A.**, Temple, C., Anderson, N., Rogalski, C., & Mentzer, K. (2025). "The Virtual Stage: Virtual Reality Integration in Effective Speaking Courses." *Information Systems Education Journal*, 23(4). [link]
6. **Patterson, A.** (2024). "Examining Text Consistency Amongst FOMC Statements and Minutes Subtext Using Deep Learning." *Issues in Information Systems*, 25(3). [link]
5. **Patterson, A.**, Frydenberg, M., & Basma, L. (2024). "Examining Generative Artificial Intelligence Adoption in Academia: a UTAUT Perspective." *Issues in Information Systems*, 25(3). [link]
4. Rruplli, E., Frydenberg, M., **Patterson, A.**, & Mentzer, K. (2024). "Examining Factors of Student AI Adoption through the Value-Based Adoption Model." *Issues in Information Systems*, 25(3). [link]
3. Mentzer, K., Frydenberg, M., & **Patterson, A.** (2024). "Are Tech Savvy Students Tech Literate? Digital and Data Literacy Skills of First-Year College Students." *Information Systems Education Journal*, 22(3). [link]
2. Grealis, T., Harmon, O., **Patterson, A.**, & Tomolonis, P. (2024). "Excel Literacy in The Classroom." *Business Education Innovation Journal*, 16(2). [link]
1. **Patterson, A.**, Mocarsky, M., Cho, J., Harmon, O., & Calvert, C. (2023). "Teaching Data Literacy and Sports Economic Fundamentals Using Fantasy Sports." *Business Education Innovation Journal*, 15(1). [link]

Presentations

20. (2026). "From Metrics to Meaning: Estimating User Feedback Using LLM-Based Evaluation." *Business Implications of Generative AI at MIT*, Cambridge, MA.
19. (2026). Discussant, "Calibrating the Curriculum: In-Demand Skills for Entry-Level Economics Majors." *American Economic Association: Conference on Teaching and Research in Economic Education*, Las Vegas, NV.
18. (2025). Discussant, "Teaching Economics More Efficiently (and Effectively) with Python." *Southern Economic Association Annual Meeting*, Tampa Bay, FL.
17. (2025). "Excel Proficiency – Assessing and Promoting." *Southern Economic Association Annual Meeting*, Tampa Bay, FL.
16. (2025). "Assessing the Alignment of FOMC Statements with Minutes using Large Language Models." *International Association for Computer Information Systems*, Clearwater, FL (virtual).
15. (2025). "From Metrics to Meaning: Estimating User Feedback Using LLM-Based Evaluation." *Amazon Economist Summit*, Seattle, WA.
14. (2025). "Onboarding Session: Setting up Coding Environment and Best Practices." *Amazon EA Summit*, Seattle, WA.
13. (2025). "Excel Literacy in The Classroom." *American Economic Association: Allied Social Science Association Annual Meeting*, San Francisco, CA.
12. (2024). "Data-Driven Decision-Making at Amazon." *Topics in Tech at Bentley University*, Waltham, MA (virtual).

11. (2024). "Examining Text Consistency amongst FOMC Statements and Minutes Subtext using Deep Learning." *International Association for Computer Information Systems*, Atlantic Beach, FL (virtual).
10. (2024). "Examining Text Consistency amongst FOMC Statements and Minutes Subtext using Deep Learning." *2nd Annual Research, Experiential, and Applied Learning Symposium*, Dudley, MA.
9. (2024). "The Rapid Rise of AI Adoption amongst College Students." *2nd Annual Research, Experiential, and Applied Learning Symposium*, Dudley, MA.
8. (2024). "Integrating VR into Effective Speaking." *2nd Annual Research, Experiential, and Applied Learning Symposium*, Dudley, MA.
7. (2023). "Excel Literacy in the Classroom and for the Future." *American Economic Association: Conference on Teaching and Research in Economic Education*, Portland, OR.
6. (2023). "Towards Measuring Data Literacy Skills." *1st Annual Research, Experiential, and Applied Learning Symposium*, Dudley, MA.
5. (2023). "Excel Literacy in the Classroom and for the Future." Poster session, *American Economic Association: Allied Social Science Association Annual Meeting*, New Orleans, LA.
4. (2022). "Using ESPN Fantasy Baseball Simulation to Teach Economic Concepts." Poster session, *American Economic Association: Allied Social Science Association Annual Meeting*, Boston, MA (virtual).
3. (2021). "Using ESPN Fantasy Baseball Simulation to Teach Economic Concepts." *Western Economic Association International 96th Annual Meeting*, Hawaii (virtual).
2. (2021). "Using ESPN Fantasy Baseball Simulation to Teach Economic Concepts." *American Economic Association: Conference on Teaching and Research in Economic Education*, Portland, OR (virtual).
1. (2020). "Using Baseball Simulation to Teach Economic Concepts." *Southern Economic Association 90th Annual Meeting*, New Orleans, LA (virtual).

Awards

IS Education Focus Area Distinguished Paper Award, ISCAP	2024
Top Conference Reviewer, ISCAP	2024
IS Education Focus Area Distinguished Paper Award, ISCAP	2023